

Play_With_Tahoe

June 1, 2026

```
[ ]: !pip install datasets
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Requirement already satisfied: datasets in /usr/local/lib/python3.12/dist-packages (4.0.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from datasets) (3.20.0)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from datasets) (2.0.2)
Requirement already satisfied: pyarrow>=15.0.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (18.1.0)
Requirement already satisfied: dill<0.3.9,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.3.8)
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Requirement already satisfied: tqdm>=4.66.3 in /usr/local/lib/python3.12/dist-packages (from datasets) (4.67.1)
Requirement already satisfied: xxhash in /usr/local/lib/python3.12/dist-packages (from datasets) (3.6.0)
Requirement already satisfied: multiprocessing<0.70.17 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.70.16)
Requirement already satisfied: fsspec<=2025.3.0,>=2023.1.0 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (2025.3.0)
Requirement already satisfied: huggingface-hub>=0.24.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.36.0)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from datasets) (25.0)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from datasets) (6.0.3)
Requirement already satisfied: aiohttp!=4.0.0a0,!4.0.0a1 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (3.13.2)
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Requirement already satisfied: certifi>=2017.4.17 in
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(2025.11.12)
Requirement already satisfied: python-dateutil>=2.8.2 in
/usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-
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packages (from pandas->datasets) (2025.2)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in
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Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-
packages (from python-dateutil>=2.8.2->pandas->datasets) (1.17.0)

```

```

[ ]: !pip install scanpy
      !pip install datasets anndata scipy pandas pubchempy

```

Requirement already satisfied: scanpy in /usr/local/lib/python3.12/dist-packages (1.11.5)

Requirement already satisfied: anndata>=0.8 in /usr/local/lib/python3.12/dist-packages (from scanpy) (0.12.6)

Requirement already satisfied: h5py>=3.7.0 in /usr/local/lib/python3.12/dist-packages (from scanpy) (3.15.1)

Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from scanpy) (1.5.2)

Requirement already satisfied: legacy-api-wrap>=1.4.1 in /usr/local/lib/python3.12/dist-packages (from scanpy) (1.5)

Requirement already satisfied: matplotlib>=3.7.5 in /usr/local/lib/python3.12/dist-packages (from scanpy) (3.10.0)

Requirement already satisfied: natsort in /usr/local/lib/python3.12/dist-packages (from scanpy) (8.4.0)

Requirement already satisfied: networkx>=2.7.1 in /usr/local/lib/python3.12/dist-packages (from scanpy) (3.6)

Requirement already satisfied: numba!=0.62.0rc1,>=0.57.1 in /usr/local/lib/python3.12/dist-packages (from scanpy) (0.60.0)

Requirement already satisfied: numpy>=1.24.1 in /usr/local/lib/python3.12/dist-packages (from scanpy) (2.0.2)

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Requirement already satisfied: session-info2 in /usr/local/lib/python3.12/dist-packages (from scanpy) (0.2.3)

Requirement already satisfied: statsmodels>=0.14.5 in /usr/local/lib/python3.12/dist-packages (from scanpy) (0.14.5)

Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from scanpy) (4.67.1)

Requirement already satisfied: typing-extensions in /usr/local/lib/python3.12/dist-packages (from scanpy) (4.15.0)

Requirement already satisfied: umap-learn>=0.5.6 in /usr/local/lib/python3.12/dist-packages (from scanpy) (0.5.9.post2)

Requirement already satisfied: array-api-compat>=1.7.1 in /usr/local/lib/python3.12/dist-packages (from anndata>=0.8->scanpy) (1.12.0)

Requirement already satisfied: zarr!=3.0.*,>=2.18.7 in /usr/local/lib/python3.12/dist-packages (from anndata>=0.8->scanpy) (3.1.5)

Requirement already satisfied: contourpy>=1.0.1 in
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Requirement already satisfied: llvmlite<0.44,>=0.43.0dev0 in
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Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-
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Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-
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Requirement already satisfied: threadpoolctl>=3.1.0 in
 /usr/local/lib/python3.12/dist-packages (from scikit-learn>=1.1.3->scanpy)
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Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-
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Requirement already satisfied: google-crc32c>=1.5 in
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Requirement already satisfied: pyyaml in /usr/local/lib/python3.12/dist-packages
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Requirement already satisfied: datasets in /usr/local/lib/python3.12/dist-
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Requirement already satisfied: anndata in /usr/local/lib/python3.12/dist-
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Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages
 (1.16.3)

Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages
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Collecting pubchempy
 Downloading pubchempy-1.0.5-py3-none-any.whl.metadata (4.3 kB)

Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from datasets) (3.20.0)

Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from datasets) (2.0.2)

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Requirement already satisfied: huggingface-hub>=0.24.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.36.0)

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Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)

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Requirement already satisfied: hf-xet<2.0.0,>=1.1.3 in

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Requirement already satisfied: multidict<7.0,>=4.5 in
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Requirement already satisfied: propcache>=0.2.0 in
/usr/local/lib/python3.12/dist-packages (from
aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (0.4.1)
Requirement already satisfied: yarll<2.0,>=1.17.0 in
/usr/local/lib/python3.12/dist-packages (from
aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets)
(1.22.0)
Downloading pubchempy-1.0.5-py3-none-any.whl (21 kB)
Installing collected packages: pubchempy

```

Successfully installed pubchempy-1.0.5

```
[ ]: from datasets import load_dataset
from scipy.sparse import csr_matrix
import anndata
import pandas as pd
import pubchempy as pc
```

```
[ ]: def create_anndata_from_generator(generator, gene_vocab, sample_size=None):
    sorted_vocab_items = sorted(gene_vocab.items())
    token_ids, gene_names = zip(*sorted_vocab_items)
    token_id_to_col_idx = {token_id: idx for idx, token_id in
↪ enumerate(token_ids)}

    data, indices, indptr = [], [], [0]
    obs_data = []

    for i, cell in enumerate(generator):
        if sample_size is not None and i >= sample_size:
            break
        genes = cell['genes']
        expressions = cell['expressions']
        if expressions[0] < 0:
            genes = genes[1:]
            expressions = expressions[1:]

        col_indices = [token_id_to_col_idx[gene] for gene in genes if gene in
↪ token_id_to_col_idx]
        valid_expressions = [expr for gene, expr in zip(genes, expressions) if
↪ gene in token_id_to_col_idx]

        data.extend(valid_expressions)
        indices.extend(col_indices)
        indptr.append(len(data))

        obs_entry = {k: v for k, v in cell.items() if k not in ['genes',
↪ 'expressions']}
        obs_data.append(obs_entry)

    expr_matrix = csr_matrix((data, indices, indptr), shape=(len(indptr) - 1,
↪ len(gene_names)))
    obs_df = pd.DataFrame(obs_data)

    adata = anndata.AnnData(X=expr_matrix, obs=obs_df)
    adata.var.index = pd.Index(gene_names, name='ensembl_id')

    return adata
```

```
[ ]: tahoe_100m_ds = load_dataset('vevotx/Tahoe-100M', streaming=True, split='train')
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94:
```

```
UserWarning:
```

```
The secret `HF_TOKEN` does not exist in your Colab secrets.
```

```
To authenticate with the Hugging Face Hub, create a token in your settings tab  
(https://huggingface.co/settings/tokens), set it as secret in your Google Colab  
and restart your session.
```

```
You will be able to reuse this secret in all of your notebooks.
```

```
Please note that authentication is recommended but still optional to access  
public models or datasets.
```

```
warnings.warn(
```

```
README.md: 0.00B [00:00, ?B/s]
```

```
Resolving data files: 0%|          | 0/3388 [00:00<?, ?it/s]
```

```
Resolving data files: 0%|          | 0/3388 [00:00<?, ?it/s]
```

```
[ ]: sample_metadata = load_dataset("vevotx/Tahoe-100M", "sample_metadata",  
    ↪split="train")
```

```
Resolving data files: 0%|          | 0/3388 [00:00<?, ?it/s]
```

```
metadata/sample_metadata.parquet: 0%|          | 0.00/65.6k [00:00<?, ?B/s]
```

```
Generating train split: 0 examples [00:00, ? examples/s]
```

```
[ ]: tahoe_100m_ds
```

```
[ ]: IterableDataset({  
    features: ['genes', 'expressions', 'drug', 'sample', 'BARCODE_SUB_LIB_ID',  
    'cell_line_id', 'moa-fine', 'canonical_smiles', 'pubchem_cid', 'plate'],  
    num_shards: 3388  
})
```

```
[ ]: gene_metadata = load_dataset("vevotx/Tahoe-100M", name="gene_metadata",  
    ↪split="train")  
gene_vocab = {entry["token_id"]: entry["ensembl_id"] for entry in gene_metadata}
```

```
Resolving data files: 0%|          | 0/3388 [00:00<?, ?it/s]
```

```
metadata/gene_metadata.parquet: 0%|          | 0.00/1.33M [00:00<?, ?B/s]
```

```
Generating train split: 0 examples [00:00, ? examples/s]
```

```
[ ]: adata = create_anndata_from_generator(tahoe_100m_ds, gene_vocab,  
    ↪sample_size=100000)  
adata
```

```
/usr/lib/python3.12/functools.py:912: ImplicitModificationWarning: Transforming
to str index.
```

```
return dispatch(args[0].__class__)(*args, **kw)
```

```
[ ]: AnnData object with n_obs × n_vars = 100000 × 62710
      obs: 'drug', 'sample', 'BARCODE_SUB_LIB_ID', 'cell_line_id', 'moa-fine',
      'canonical_smiles', 'pubchem_cid', 'plate'
```

```
[ ]: adata.obs.shape
```

```
[ ]: (100000, 8)
```

```
[ ]: sample_metadata = load_dataset("vevotx/Tahoe-100M", "sample_metadata",
      ↪split="train").to_pandas()
adata.obs = pd.merge(adata.obs, sample_metadata.drop(columns=["drug", "plate"]),
      ↪on="sample")
adata.obs.head()
```

```
Resolving data files: 0%|          | 0/3388 [00:00<?, ?it/s]
```

```
/usr/lib/python3.12/functools.py:912: ImplicitModificationWarning: Transforming
to str index.
```

```
return dispatch(args[0].__class__)(*args, **kw)
```

```
[ ]:
      drug      sample  BARCODE_SUB_LIB_ID  cell_line_id  moa-fine  \
0  8-Hydroxyquinoline  smp_1783  01_001_052-lib_1105    CVCL_0480  unclear
1  8-Hydroxyquinoline  smp_1783  01_001_105-lib_1105    CVCL_0546  unclear
2  8-Hydroxyquinoline  smp_1783  01_001_165-lib_1105    CVCL_1717  unclear
3  8-Hydroxyquinoline  smp_1783  01_003_094-lib_1105    CVCL_1717  unclear
4  8-Hydroxyquinoline  smp_1783  01_003_164-lib_1105    CVCL_1056  unclear
```

```

      canonical_smiles  pubchem_cid  plate  mean_gene_count  \
0  C1=CC2=C(C(=C1)O)N=CC=C2      1923.0  plate4      1478.268171
1  C1=CC2=C(C(=C1)O)N=CC=C2      1923.0  plate4      1478.268171
2  C1=CC2=C(C(=C1)O)N=CC=C2      1923.0  plate4      1478.268171
3  C1=CC2=C(C(=C1)O)N=CC=C2      1923.0  plate4      1478.268171
4  C1=CC2=C(C(=C1)O)N=CC=C2      1923.0  plate4      1478.268171
```

```

      mean_tscp_count  mean_mread_count  mean_pcmt_mito  \
0      2341.339094      2738.463797      0.023783
1      2341.339094      2738.463797      0.023783
2      2341.339094      2738.463797      0.023783
3      2341.339094      2738.463797      0.023783
4      2341.339094      2738.463797      0.023783
```

```

      drugname_drugconc
0  [('8-Hydroxyquinoline', 0.05, 'uM')]
1  [('8-Hydroxyquinoline', 0.05, 'uM')]
```

```
2 [('8-Hydroxyquinoline', 0.05, 'uM')]
3 [('8-Hydroxyquinoline', 0.05, 'uM')]
4 [('8-Hydroxyquinoline', 0.05, 'uM')]
```

```
[ ]: drug_metadata = load_dataset("vevotx/Tahoe-100M", "drug_metadata",
    ↪split="train").to_pandas()
adata.obs = pd.merge(adata.obs, drug_metadata.
    ↪drop(columns=["canonical_smiles", "pubchem_cid", "moa-fine"]), on="drug",
    ↪how='left')
adata.obs.head()
```

Resolving data files: 0%| | 0/3388 [00:00<?, ?it/s]

/usr/lib/python3.12/functools.py:912: ImplicitModificationWarning: Transforming to str index.

```
return dispatch(args[0].__class__)(*args, **kw)
```

```
[ ]:
      drug      sample  BARCODE_SUB_LIB_ID  cell_line_id  moa-fine  \
0  8-Hydroxyquinoline  smp_1783  01_001_052-lib_1105  CVCL_0480  unclear
1  8-Hydroxyquinoline  smp_1783  01_001_105-lib_1105  CVCL_0546  unclear
2  8-Hydroxyquinoline  smp_1783  01_001_165-lib_1105  CVCL_1717  unclear
3  8-Hydroxyquinoline  smp_1783  01_003_094-lib_1105  CVCL_1717  unclear
4  8-Hydroxyquinoline  smp_1783  01_003_164-lib_1105  CVCL_1056  unclear
```

```
      canonical_smiles  pubchem_cid  plate  mean_gene_count  \
0  C1=CC2=C(C(=C1)O)N=CC=C2      1923.0  plate4      1478.268171
1  C1=CC2=C(C(=C1)O)N=CC=C2      1923.0  plate4      1478.268171
2  C1=CC2=C(C(=C1)O)N=CC=C2      1923.0  plate4      1478.268171
3  C1=CC2=C(C(=C1)O)N=CC=C2      1923.0  plate4      1478.268171
4  C1=CC2=C(C(=C1)O)N=CC=C2      1923.0  plate4      1478.268171
```

```
      mean_tscp_count  mean_mread_count  mean_pcmt_mito  \
0      2341.339094      2738.463797      0.023783
1      2341.339094      2738.463797      0.023783
2      2341.339094      2738.463797      0.023783
3      2341.339094      2738.463797      0.023783
4      2341.339094      2738.463797      0.023783
```

```
      drugname_drugconc  targets  moa-broad  human-approved  \
0  [('8-Hydroxyquinoline', 0.05, 'uM')]  None  unclear  no
1  [('8-Hydroxyquinoline', 0.05, 'uM')]  None  unclear  no
2  [('8-Hydroxyquinoline', 0.05, 'uM')]  None  unclear  no
3  [('8-Hydroxyquinoline', 0.05, 'uM')]  None  unclear  no
4  [('8-Hydroxyquinoline', 0.05, 'uM')]  None  unclear  no
```

```
      clinical-trials  gpt-notes-approval
0      yes  Used in some clinical trial formulations, not ...
```

```

1         yes Used in some clinical trial formulations, not ...
2         yes Used in some clinical trial formulations, not ...
3         yes Used in some clinical trial formulations, not ...
4         yes Used in some clinical trial formulations, not ...

```

```

[ ]: cell_line_metadata = load_dataset("vevotx/Tahoe-100M", "cell_line_metadata",
    ↪split="train").to_pandas()
cell_line_metadata.head(1)

```

```
Resolving data files:  0%|          | 0/3388 [00:00<?, ?it/s]
```

```

[ ]:   cell_name Cell_ID_DepMap Cell_ID_Cellosaur Organ Driver_Gene_Symbol \
0      A549      ACH-000681      CVCL_0023 Lung      CDKN2A

      Driver_VarZyg Driver_VarType Driver_ProtEffect_or_CdnaEffect \
0           Hom      Deletion                                DEL

      Driver_Mech_InferDM Driver_GeneType_DM
0                    LoF      Suppressor

```

```
[ ]: cell_line_metadata.shape
```

```
[ ]: (1000, 10)
```

```
[ ]: adata.obs.shape
```

```
[ ]: (100000, 18)
```

```
[ ]: adata.obs.head(1)
```

```

[ ]:   drug      sample  BARCODE_SUB_LIB_ID cell_line_id moa-fine \
0  8-Hydroxyquinoline  smp_1783  01_001_052-lib_1105  CVCL_0480  unclear

      canonical_smiles pubchem_cid  plate  mean_gene_count \
0  C1=CC2=C(C(=C1)O)N=CC=C2      1923.0  plate4      1478.268171

      mean_tscp_count  mean_mread_count  mean_pcmt_mito \
0      2341.339094      2738.463797      0.023783

      drugname_drugconc targets moa-broad human-approved \
0  [('8-Hydroxyquinoline', 0.05, 'uM')]  None  unclear      no

      clinical-trials                                gpt-notes-approval
0      yes Used in some clinical trial formulations, not ...

```

```

[ ]: cell_line_metadata_unique = cell_line_metadata.
    ↪drop_duplicates(subset=['Cell_ID_Cellosaur'])

```

```
adata.obs = pd.merge(adata.obs, cell_line_metadata_unique,
    ↪left_on='cell_line_id' , right_on='Cell_ID_Cellosaur', how='left')
adata.obs.head()
```

```
/usr/lib/python3.12/functools.py:912: ImplicitModificationWarning: Transforming
to str index.
```

```
return dispatch(args[0].__class__)(*args, **kw)
```

```
[ ]:      drug      sample  BARCODE_SUB_LIB_ID  cell_line_id  moa-fine  \
0  8-Hydroxyquinoline  smp_1783  01_001_052-lib_1105    CVCL_0480  unclear
1  8-Hydroxyquinoline  smp_1783  01_001_105-lib_1105    CVCL_0546  unclear
2  8-Hydroxyquinoline  smp_1783  01_001_165-lib_1105    CVCL_1717  unclear
3  8-Hydroxyquinoline  smp_1783  01_003_094-lib_1105    CVCL_1717  unclear
4  8-Hydroxyquinoline  smp_1783  01_003_164-lib_1105    CVCL_1056  unclear
```

```
      canonical_smiles  pubchem_cid  plate  mean_gene_count  \
0  C1=CC2=C(C(=C1)O)N=CC=C2      1923.0  plate4      1478.268171
1  C1=CC2=C(C(=C1)O)N=CC=C2      1923.0  plate4      1478.268171
2  C1=CC2=C(C(=C1)O)N=CC=C2      1923.0  plate4      1478.268171
3  C1=CC2=C(C(=C1)O)N=CC=C2      1923.0  plate4      1478.268171
4  C1=CC2=C(C(=C1)O)N=CC=C2      1923.0  plate4      1478.268171
```

```
      mean_tscp_count  ...  cell_name  Cell_ID_DepMap  Cell_ID_Cellosaur  \
0      2341.339094  ...    PANC-1      ACH-000164      CVCL_0480
1      2341.339094  ...    SW480      ACH-000842      CVCL_0546
2      2341.339094  ...    SW1417      ACH-000236      CVCL_1717
3      2341.339094  ...    SW1417      ACH-000236      CVCL_1717
4      2341.339094  ...     A498      ACH-000555      CVCL_1056
```

```
      Organ  Driver_Gene_Symbol  Driver_VarZyg  Driver_VarType  \
0  Pancreas      AKT2      None      Gain
1   Bowel      APC      Hom      Stopgain
2   Bowel      APC      Hom      Stopgain
3   Bowel      APC      Hom      Stopgain
4  Kidney      APC      Het      Missense
```

```
      Driver_ProtEffect_or_CdnaEffect  Driver_Mech_InferDM  Driver_GeneType_DM
0      GAIN      GoF      Oncogene
1  p.Q1338*      LoF      Suppressor
2  p.R1450*      LoF      Suppressor
3  p.R1450*      LoF      Suppressor
4  p.C1270R      LoF      Suppressor
```

```
[5 rows x 28 columns]
```

```
[ ]: cell_line_metadata.shape
```

```
[ ]: (1000, 10)
```

```
[ ]: adata
```

```
[ ]: AnnData object with n_obs × n_vars = 1000 × 62710
      obs: 'drug', 'sample', 'BARCODE_SUB_LIB_ID', 'cell_line_id', 'moa-fine',
      'canonical_smiles', 'pubchem_cid', 'plate', 'mean_gene_count',
      'mean_tscp_count', 'mean_mread_count', 'mean_pcmt_mito', 'drugname_drugconc',
      'targets', 'moa-broad', 'human-approved', 'clinical-trials', 'gpt-notes-
      approval', 'cell_name', 'Cell_ID_DepMap', 'Cell_ID_Cellosaur', 'Organ',
      'Driver_Gene_Symbol', 'Driver_VarZyg', 'Driver_VarType',
      'Driver_ProtEffect_or_CdnaEffect', 'Driver_Mech_InferDM', 'Driver_GeneType_DM'
```

```
[ ]: adata.obs.head()
```

```
[ ]:
      drug      sample  BARCODE_SUB_LIB_ID  cell_line_id  moa-fine  \
0  8-Hydroxyquinoline  smp_1783  01_001_052-lib_1105    CVCL_0480  unclear
1  8-Hydroxyquinoline  smp_1783  01_001_105-lib_1105    CVCL_0546  unclear
2  8-Hydroxyquinoline  smp_1783  01_001_165-lib_1105    CVCL_1717  unclear
3  8-Hydroxyquinoline  smp_1783  01_003_094-lib_1105    CVCL_1717  unclear
4  8-Hydroxyquinoline  smp_1783  01_003_164-lib_1105    CVCL_1056  unclear
```

```

      canonical_smiles  pubchem_cid  plate  mean_gene_count  \
0  C1=CC2=C(C(=C1)O)N=CC=C2    1923.0  plate4    1478.268171
1  C1=CC2=C(C(=C1)O)N=CC=C2    1923.0  plate4    1478.268171
2  C1=CC2=C(C(=C1)O)N=CC=C2    1923.0  plate4    1478.268171
3  C1=CC2=C(C(=C1)O)N=CC=C2    1923.0  plate4    1478.268171
4  C1=CC2=C(C(=C1)O)N=CC=C2    1923.0  plate4    1478.268171
```

```

      mean_tscp_count  ...  cell_name  Cell_ID_DepMap  Cell_ID_Cellosaur  \
0      2341.339094  ...    PANC-1    ACH-000164    CVCL_0480
1      2341.339094  ...    SW480    ACH-000842    CVCL_0546
2      2341.339094  ...    SW1417    ACH-000236    CVCL_1717
3      2341.339094  ...    SW1417    ACH-000236    CVCL_1717
4      2341.339094  ...    A498    ACH-000555    CVCL_1056
```

```

      Organ  Driver_Gene_Symbol  Driver_VarZyg  Driver_VarType  \
0  Pancreas          AKT2          None          Gain
1   Bowel           APC          Hom    Stopgain
2   Bowel           APC          Hom    Stopgain
3   Bowel           APC          Hom    Stopgain
4  Kidney           APC          Het    Missense
```

```

      Driver_ProtEffect_or_CdnaEffect  Driver_Mech_InferDM  Driver_GeneType_DM
0              GAIN              GoF          Oncogene
1      p.Q1338*              LoF          Suppressor
2      p.R1450*              LoF          Suppressor
```

3	p.R1450*	LoF	Suppressor
4	p.C1270R	LoF	Suppressor

[5 rows x 28 columns]

```
[ ]: import pandas as pd
import gzip

def parse_geo_metadata(file_path):
    # Initialize dictionary to store data
    metadata = {
        "geo_accession": [],
        "sample_title": [],
        "sex": [],
        "condition": []
    }

    # Logic to handle different file encodings/types if needed
    try:
        f = open(file_path, 'r')
    except:
        f = gzip.open(file_path, 'rt')

    with f:
        for line in f:
            # Clean up line
            line = line.strip().replace('"', '')

            # Extract GSM Accessions
            if line.startswith('!Sample_geo_accession'):
                # Skip the first element which is the header "!"
                ↪Sample_geo_accession"
                metadata["geo_accession"] = line.split('\t')[1:]

            # Extract Sample Titles (e.g. SCP3412)
            elif line.startswith('!Sample_title'):
                metadata["sample_title"] = line.split('\t')[1:]

            # Extract Characteristics (Sex and Condition)
            elif line.startswith('!Sample_characteristics_ch1'):
                values = line.split('\t')[1:]

            # Check what kind of characteristic this row is
            if "genotype:" in values[0]:
                # Clean the string to keep only 'male' or 'female'
                metadata["sex"] = [x.replace('genotype: ', '') for x in ↪
                ↪values]
```

```

        elif "treatment:" in values[0]:
            # Clean the string to keep only 'Preeclampsia' or 'Control'
            metadata["condition"] = [x.replace('treatment: ', '') for x
in values]

    # Create DataFrame
    df = pd.DataFrame(metadata)

    return df

```

	geo_accession	sample_title	sex	condition
0	GSM9210996	SCP3412	male	Preeclampsia
1	GSM9210997	SCP3628	female	Control
2	GSM9210998	SCP3660	female	Control
3	GSM9210999	SCP3780	female	Control
4	GSM9211000	SCP3843	female	Control

```

[ ]: # Usage
# Save your file as 'GSE306864_series_matrix.txt' in your folder
metadata_df = parse_geo_metadata('GSE306864_series_matrix.txt')

# Inspect the cleaned data
print(metadata_df.head())

# Save to CSV for your pipeline
metadata_df.to_csv("GSE306864_metadata_cleaned.csv", index=False)

```

	geo_accession	sample_title	sex	condition
0	GSM9210996	SCP3412	male	Preeclampsia
1	GSM9210997	SCP3628	female	Control
2	GSM9210998	SCP3660	female	Control
3	GSM9210999	SCP3780	female	Control
4	GSM9211000	SCP3843	female	Control

```

[ ]: metadata_df

```

```

[ ]:

```

	geo_accession	sample_title	sex	condition
0	GSM9210996	SCP3412	male	Preeclampsia
1	GSM9210997	SCP3628	female	Control
2	GSM9210998	SCP3660	female	Control
3	GSM9210999	SCP3780	female	Control
4	GSM9211000	SCP3843	female	Control
5	GSM9211001	SCP3877	female	Control
6	GSM9211002	SCP3929	male	Control
7	GSM9211003	SCP3954	male	Control
8	GSM9211004	SCP3962	male	Preeclampsia

9	GSM9211005	SCP3992	female	Control
10	GSM9211006	SCP4059	male	Control
11	GSM9211007	SCP4073	female	Control
12	GSM9211008	SCP4139	female	Control
13	GSM9211009	SCP4154	female	Preeclampsia
14	GSM9211010	SCP4157	male	Preeclampsia
15	GSM9211011	SCP4164	female	Preeclampsia
16	GSM9211012	SCP4196	male	Control
17	GSM9211013	SCP4319	male	Control
18	GSM9211014	SCP4378	female	Control
19	GSM9211015	SCP4536	female	Control
20	GSM9211016	SCP4538	female	Control
21	GSM9211017	SCP4578	male	Control
22	GSM9211018	SCP4706	female	Control
23	GSM9211019	SCP4726	female	Preeclampsia
24	GSM9211020	SCP4733	female	Control
25	GSM9211021	SCP4748	female	Control
26	GSM9211022	SCP4809	male	Control
27	GSM9211023	SCP4913	male	Control
28	GSM9211024	STP0016	female	Control
29	GSM9211025	STP0023	female	Control
30	GSM9211026	STP0105	male	Control
31	GSM9211027	STP0148	male	Preeclampsia
32	GSM9211028	STP0230	female	Preeclampsia
33	GSM9211029	STP0275	female	Control
34	GSM9211030	STP0282	female	Preeclampsia
35	GSM9211031	STP0378	male	Control
36	GSM9211032	STP0582	female	Control
37	GSM9211033	STP0583	male	Control
38	GSM9211034	STP0593	female	Preeclampsia
39	GSM9211035	STP0596	male	Control
40	GSM9211036	STP0636	female	Control
41	GSM9211037	STP0677	female	Preeclampsia
42	GSM9211038	STP0692	female	Control
43	GSM9211039	STP0761	female	Control
44	GSM9211040	STP0788	female	Control
45	GSM9211041	STP0793	female	Preeclampsia
46	GSM9211042	STP0795	male	Control
47	GSM9211043	STP0852	female	Control
48	GSM9211044	STP0867	female	Control
49	GSM9211045	STP0888	female	Control
50	GSM9211046	STP0903	female	Control
51	GSM9211047	STP0912	female	Control
52	GSM9211048	STP0932	male	Control
53	GSM9211049	STP0944	male	Preeclampsia
54	GSM9211050	STP1002	male	Control
55	GSM9211051	STP1082	female	Control

56	GSM9211052	STP1087	female	Control
57	GSM9211053	STP1206	male	Control

```
[ ]: metadata_df.to_csv('GSE306864_series_matrix_cleaned.csv', index=None)
```

```
[ ]:
```